AI-Powered Report Assistant Platform

Natural Language → SQL → Insights

Executive Summary

The AI-Powered SQL Query Assistant Platform provides a smart and user-friendly interface that allows employees to request information in natural language. The system understands the request, retrieves data from the selected database, and returns meaningful ready-to-use insights such as reports, charts, summaries, tables, or email drafts.

This solution reduces IT workload, accelerates decision-making, and centralizes secure access to organizational data.

Application Overview

User Interface

A simple, modern one-page interface featuring:
- A sidebar showing all applications assigned to the logged-in user.
- A recent activity panel showing the user's last 10 requests.
- A form with two required fields: Subject and Details.

Workflow Summary

User enters a request → System understands the meaning → AI generates the correct SQL → System retrieves data → AI produces final output → User receives insights.

Features

• Natural-language request interpretation
• Automatic SQL generation
• Secure data retrieval from multiple databases
• AI-generated summaries, reports, charts, tables, and email drafts
• Compatible with Gemini AI, OpenAI, Claude, Llama, Mistral, and Cohere
• Supports on-premise LLM hosting

Visual Overview

System Flow:
User Request → AI Interpretation → SQL Generation → Database Execution → AI Output → Insight Delivered

Value to the Organization

For Business Teams:
- Instant access to insights

- No SQL knowledge required
- Faster decisions
- Self-service analytics

For IT Teams:
- Fewer manual requests
- More time for strategic tasks
- Secure, centralized data access
- Scalable foundation

Technology Stack

• ASP.NET MVC
• .NET Core Web API
• Oracle 19c (primary database)
• Optional SQL Server / MySQL
• AI provider API key
• Secure configuration storage

Roadmap

Short-Term:
- Dashboards and charts
- Export to PDF/Excel
- Multi-turn AI reasoning

Medium-Term:
- On-premises AI hosting
- Data masking
- Monitoring pages for administrators
- Support more database engines to cover all SAA applications.

Long-Term:
- BI integration
- Analytics automation (scheduling auto-generate and send to emails, etc.)
- AI assistant marketplace

Suggestions

• Add SQL safety validation
• Enable caching to reduce cost
• Support multiple AI models
• Plan for secure on-prem LLM clusters.

Conclusion

This platform revolutionizes data access by allowing users to obtain insights instantly without technical skills. It empowers business users while reducing IT workload and preparing the organization for future AI advancements.